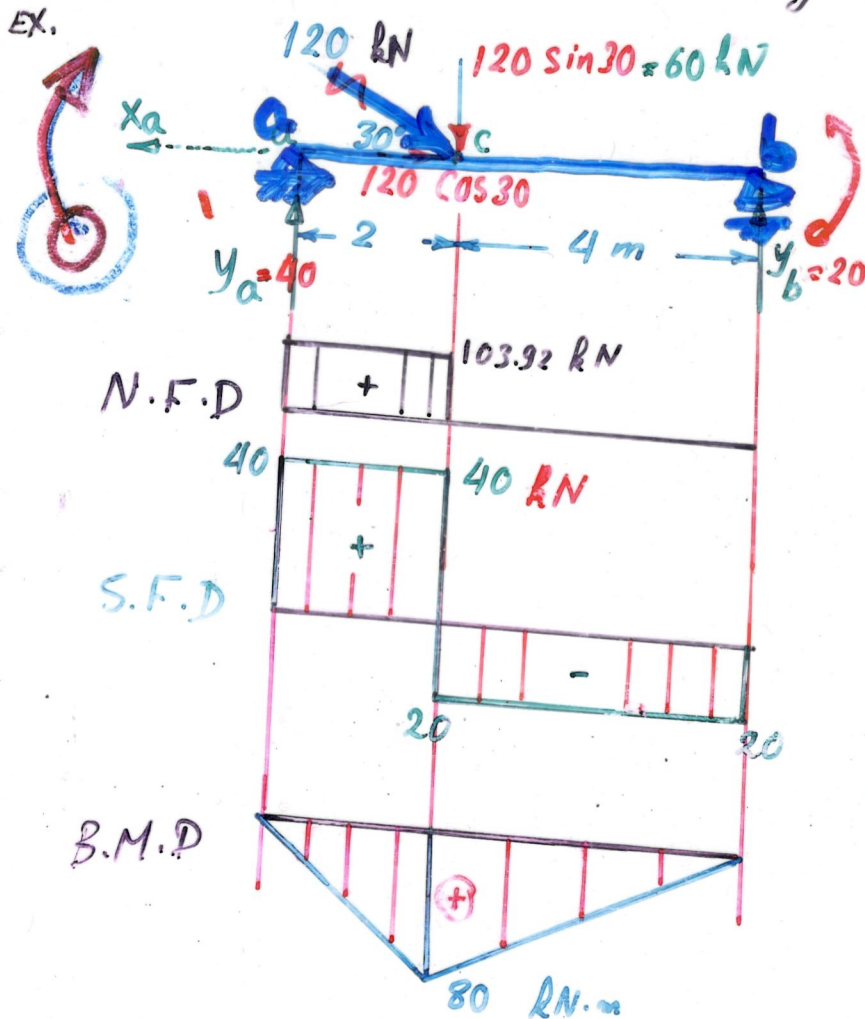


Calculate the reactions for the following statically determinate structures. Then draw the normal force, shearing force and bending moment diagrams.



$$120 \cos 30 = 103.92 \text{ kN}$$

$$120 \sin 30 = 60 \text{ kN}$$

$$\sum M_a = 0$$

$$60 \times 4 = y_a \times 6$$

$$y_a = 40 \text{ kN}$$

$$\sum y = 0$$

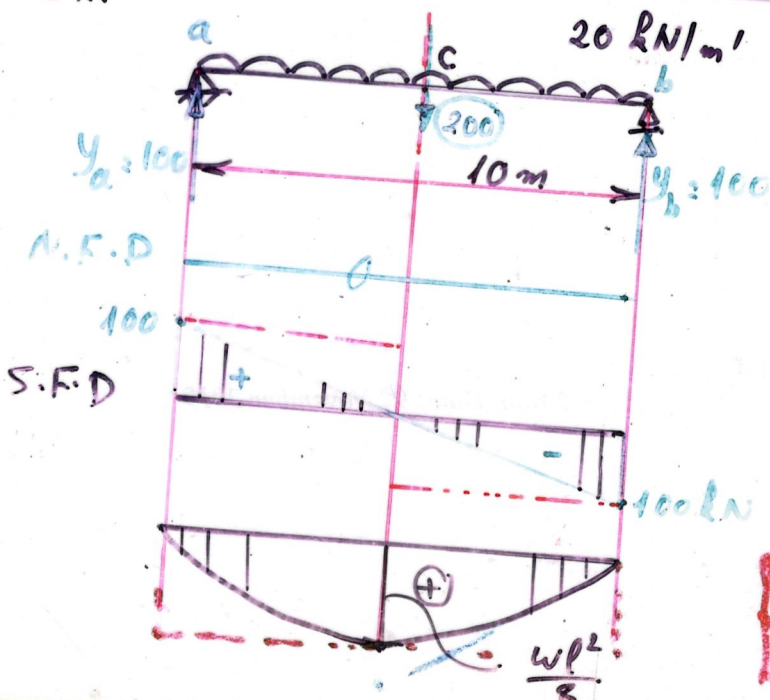
$$y_b = 60 - 40 = 20 \text{ kN}$$

$$\sum x = 0$$

$$x_a = 103.92 \text{ kN}$$

$$M_c = 20 \times 4 = 80 \text{ kN.m}$$

EX.



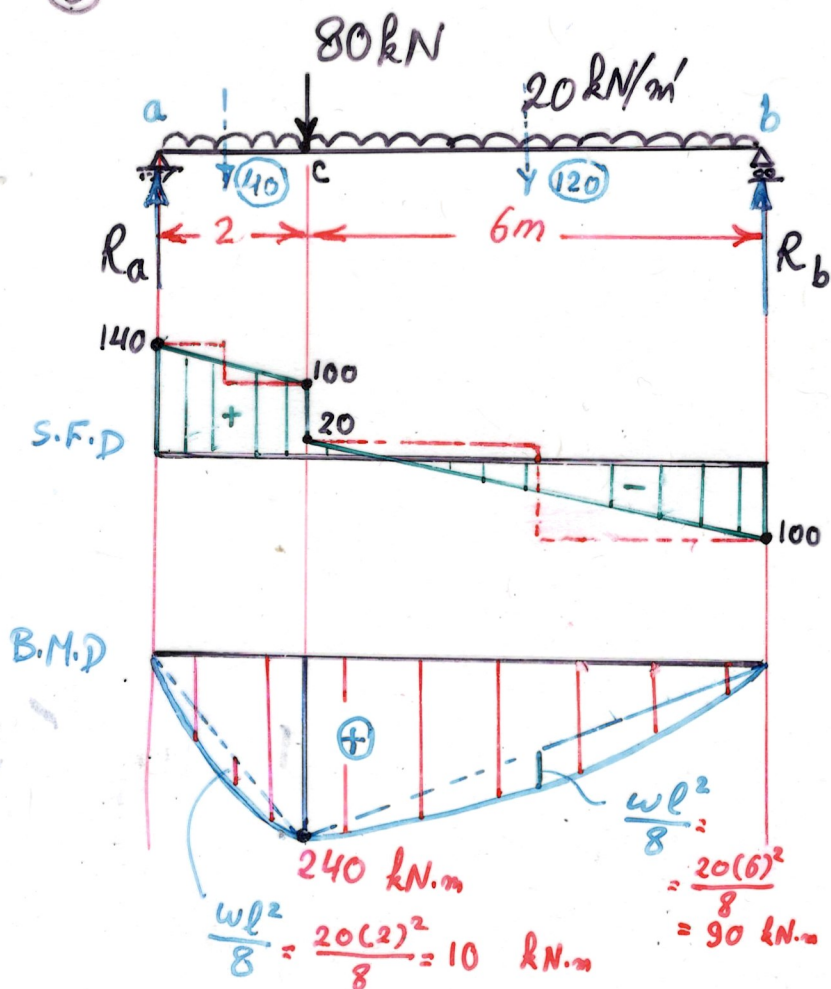
$$y_a = y_b = \frac{200}{2} = 100 \text{ kN}$$

$$\text{max B.M at } c = \frac{wl^2}{8}$$

$$= \frac{20(10)^2}{8}$$

$$= 250 \text{ kN.m}$$

③



$M \rightarrow +ve$

$$\sum M_a = 0$$

$$R_b = \frac{40 \times 1 + 80 \times 2 + 120 \times 5}{8}$$

$$R_b = 100 \text{ kN}$$

$$\sum Y = 0$$

$$R_a = 80 + 40 + 120 - 100$$

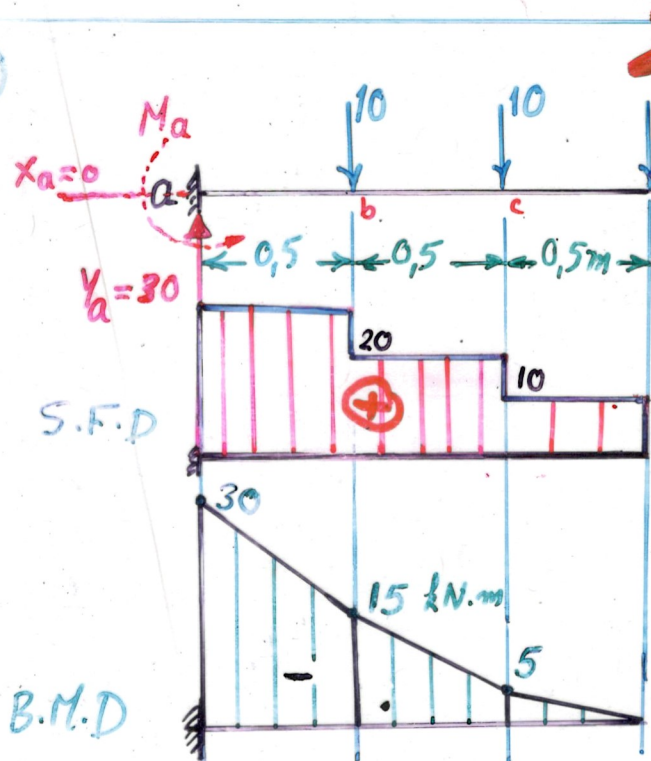
$$R_a = 140 \text{ kN}$$

B.M. at c

$$M = 140 \times 2 - 40 \times 1$$

$$= 240 \text{ kN.m}$$

④



$$\sum X = 0$$

$$\therefore X_a = 0$$

$$\sum Y = 0$$

$$Y_a = 10 + 10 + 10$$

$$= 30 \text{ kN}$$

$$M_d = 0$$

$$M_c = -10 \times 0.5 = 5 \text{ kN.m}$$

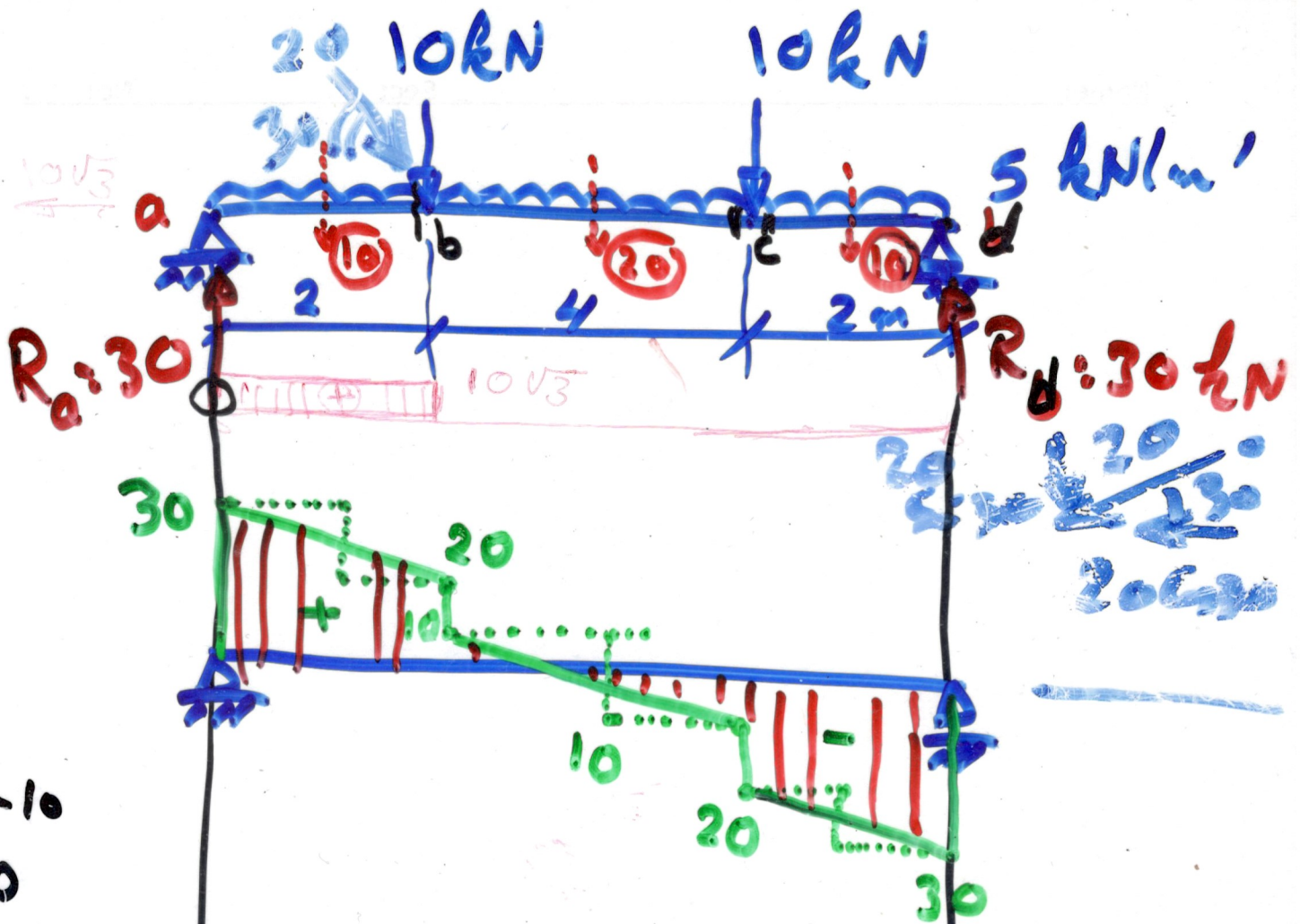
$$M_b = -10 \times 1 - 10 \times 0.5 = 15 \text{ kN.m}$$

$$M_a = -10 \times 1.5 - 10 \times 1 - 10 \times 0.5$$

$$= -30 \text{ kN.m} \equiv 30 \text{ kN.m}$$

20 kN 30

N.F.D



$$\frac{wL^2}{8}$$

$$\frac{5(2)^2}{8}$$

$$2.5 \text{ kN.m}$$

$$\frac{5(4)^2}{8}$$

$$10 \text{ kN.m}$$

B.M.D.

2.5 kN.m

50 kN.m

50 kN.m